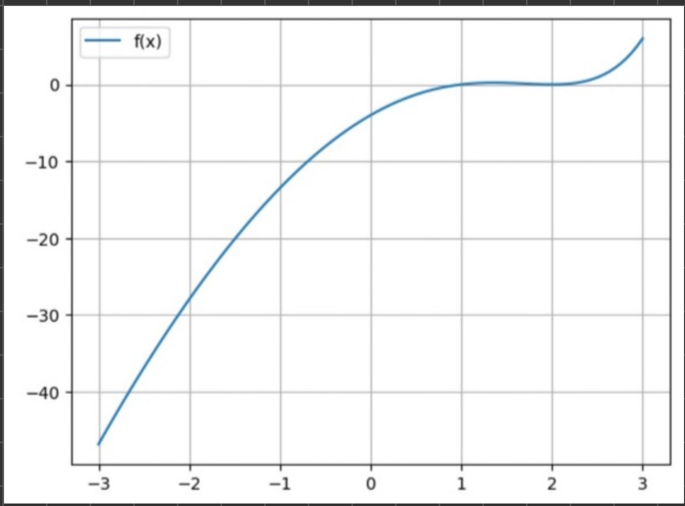
Prob. 1 – Bisection method

Ex 4 – K.P. Lind

$$f(x) = (1-x)\,2^{x+2} + (2^x - 2)\,x^2 + 8(x-1)$$

a)



b)

$$f(x) = (1-x)\,2^{x+2} + (2^x - 2)\,x^2 + 8(x-1)$$

$a = 0$ & $b = 3$: $f(0) = -4$ $\quad f(3) = 6 \Rightarrow f(0)f(3) = -24 < 0$

$c = \frac{a+b}{2} = \frac{0+3}{2} = \frac{3}{2} \Rightarrow f(c) = 0{,}207 \Rightarrow f(a)f(c) < 0$

$a = 0$ & $b = \frac{3}{2} \Rightarrow c = \frac{a+b}{2} = \frac{0+\frac{3}{2}}{2} = \frac{3}{4} \Rightarrow f(c) = -0{,}497 \Rightarrow f(0)f(\tfrac{3}{4}) > 0$

$a = \frac{3}{4}$ & $b = \frac{3}{2} \Rightarrow c = \frac{\frac{3}{4}+\frac{3}{2}}{2} = \frac{9}{8} \Rightarrow f(c) = 0{,}139 \Rightarrow f(\tfrac{3}{4})f(\tfrac{9}{8}) < 0$

$a = \frac{3}{4}$ & $b = \frac{9}{8} \Rightarrow c = \frac{\frac{3}{4}+\frac{9}{8}}{2} = \frac{15}{16} \Rightarrow f(c) = -0{,}096$ → Errorbound: 10^{-1}

c & d)

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bisection(f, -3, 3, tol=1.0e-3, max_iter = 100, verbose=True)

k=  0, a=-3.000000, b=3.000000, c=  0.000000, f(c)=-4.000e+00
k=  1, a=0.000000, b=3.000000, c=  1.500000, f(c)= 2.071e-01
k=  2, a=0.000000, b=1.500000, c=  0.750000, f(c)=-4.972e-01
k=  3, a=0.750000, b=1.500000, c=  1.125000, f(c)= 1.386e-01
k=  4, a=0.750000, b=1.125000, c=  0.937500, f(c)=-9.572e-02
k=  5, a=0.937500, b=1.125000, c=  1.031250, f(c)= 4.110e-02
k=  6, a=0.937500, b=1.031250, c=  0.984375, f(c)=-2.222e-02
k=  7, a=0.984375, b=1.031250, c=  1.007812, f(c)= 1.069e-02
k=  8, a=0.984375, b=1.007812, c=  0.996094, f(c)=-5.450e-03
k=  9, a=0.996094, b=1.007812, c=  1.001953, f(c)= 2.699e-03
k= 10, a=0.996094, b=1.001953, c=  0.999023, f(c)=-1.356e-03
k= 11, a=0.999023, b=1.001953, c=  1.000488, f(c)= 6.764e-04
k= 12, a=0.999023, b=1.000488, c=  0.999756, f(c)=-3.386e-04
The point c_k is close to r (less than 0.000732).

(0.999755859375, 12)
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12 iterations to guarantee an error smaller than 10^{-3}.

Prob. 2 – Radius of Convergence

$f(x) = x\log(x) - x$

a) $\left|\frac{f''(y)}{f'(x)}\right| \le 2M(a,b)$, $\forall xy \in [a,b]$, $0 < a < b < \infty$

$f'(x) = x \cdot \frac{1}{x} + 1\log(x) - 1 = \log(x)$

$f''(y) = \frac{1}{y}$

$\left|\frac{f''(y)}{f'(x)}\right| = \left|\frac{1}{y\log(x)}\right| \le 2M(a,b) \Rightarrow \left|\frac{1}{2y\log(x)}\right| \le M(a,b)$

If $a = 1$ $\log(a) = 0 \Rightarrow$ error

If $a < 1$ $M(a,b) \Rightarrow \infty$

If $a > 1$ $M(a,b) = \frac{1}{2a\log(a)}$

b)

Prob. 3 – Periodic functions

a) Every constant function is always periodic, with no fundamental period.

Counterex: $f(x) = 1$

b) • $f(x) = \sin(x+2) \Rightarrow T = 2\pi$

Period of trig. func. is 2π. If a constant is added, subtracted, multiplied or divided, period stays the same

• $f(x) = \pi\cos(\pi x) \Rightarrow T = 2$

If $f(x)$ is a func. with period P, then $f(ax)$, $a \in \mathbb{R} \neq 0$, is periodic with period $\frac{P}{|a|} = \frac{2\pi}{|a|}$.

• $f(x) = \cos\left(\frac{2\pi}{m+1}x\right) + \sin\left(\frac{2\pi}{n+1}x\right)$, $n, m \in \mathbb{N} \setminus \{1\} \Rightarrow T = \mathrm{LCM}(m+1, n+1)$

Prob. 4

$f(-x) = -f(x) \qquad g(-x) = g(x)$

a) $h(x) = f(x)g(x)$

$h(-x) = f(-x)g(-x)$

$h(-x) = -f(x)g(x)$

$h(-x) = -h(x) \Rightarrow$ odd

b) $h(x) = f^2(x)$

$h(x) = f(x)f(x)$

$h(-x) = f(-x)f(-x)$

$h(-x) = (-f(x))(-f(x))$

$h(-x) = f(x)f(x)$

$h(-x) = f^2(x)$

$h(-x) = h(x) \Rightarrow$ even

c) $\int_{-L}^{L} f(x)\,dx = 0 \quad , \quad f(-x) = -f(x)$

$\int_{-L}^{L} f(x)\,dx = \int_{-a}^{0} f(x)\,dx + \int_{0}^{a} f(x)\,dx$

Then we have to prove that

$\int_{-a}^{0} f(x)\,dx = -\int_{0}^{a} f(x)\,dx$

$\int_{\beta(a)}^{\beta(0)} f(x)\,dx = \int_{a}^{0} f(-u)(-1)\,du$

$= \int_{0}^{a} f(-u)\,du$

$= \int_{0}^{a} f(-x)\,dx$

$= -\int_{0}^{a} f(x)\,dx \Rightarrow$ since $f(x)$ is an odd function

d) $(f \circ g)(x) = f(g(x))$

$(f \circ g)(-x) = f(g(-x))$

$(f \circ g)(-x) = f(g(x))$

$(f \circ g)(-x) = (f \circ g)(x) \Rightarrow$ even

$(g \circ f)(x) = g(f(x))$

$(g \circ f)(-x) = g(f(-x))$

$(g \circ f)(-x) = g(-f(x))$

$(g \circ f)(-x) = g(f(x))$

$(g \circ f)(-x) = (g \circ f)(x) \Rightarrow$ even

e) $\int_{-L}^{L} f(x)\,dx = 2\int_{0}^{L} f(x)\,dx \quad , \quad f(-x) = f(x)$

$\int_{-L}^{L} f(x)\,dx = \int_{-a}^{0} f(x)\,dx + \int_{0}^{a} f(x)\,dx$

Then we have to prove that

$\int_{a}^{0} f(x)\,dx = \int_{0}^{a} f(x)\,dx$

$\int_{\beta(a)}^{\beta(0)} f(x)\,dx = \int_{a}^{0} f(-u)(-1)\,du$

$= \int_{0}^{a} f(-u)\,du$

$= \int_{0}^{a} f(-x)\,dx$

$= \int_{0}^{a} f(x)\,dx \Rightarrow$ since $f(x)$ is an even function

Prob. 5 – Trigonometric interpolation

$$y = f(x) = 2 + \sin(x) + \sin(2x)$$

	y_0	y_1	y_2
y	2	$2+\frac{\sqrt{3}}{4}$	$2+\frac{\sqrt{3}}{4}$
x	0	$\pi/6$	$\pi/3$
	x_0	x_1	x_2

a) $$p(x) = \alpha_1 + \alpha_2 \sin(x) + \alpha_3 \sin(2x)$$

$$f(x_0) = p(x_0) = \alpha_1 + \alpha_2 \sin(x_0) + \alpha_3 \sin(2x_0)$$

$$f(x_1) = p(x_1) = \alpha_1 + \alpha_2 \sin(x_1) + \alpha_3 \sin(2x_1)$$

$$f(x_2) = p(x_2) = \alpha_1 + \alpha_2 \sin(x_2) + \alpha_3 \sin(2x_2)$$

$$A = \begin{array}{c} \begin{array}{ccc} \alpha_0 & \alpha_1 & \alpha_2 \end{array} \\ \begin{bmatrix} 1 & \sin(x_0) & \sin(2x_0) \\ 1 & \sin(x_1) & \sin(2x_1) \\ 1 & \sin(x_2) & \sin(2x_2) \end{bmatrix} \end{array}, \qquad b = \begin{bmatrix} f(x_0) \\ f(x_1) \\ f(x_2) \end{bmatrix}$$

$$\begin{array}{c} \begin{array}{cccc} \alpha_0 & \alpha_1 & \alpha_2 & \end{array} \\ \left[\begin{array}{ccc|c} 1 & \sin(0) & \sin(0) & 2 \\ 1 & \sin(\pi/6) & \sin(\pi/3) & 2+\frac{\sqrt{3}}{4} \\ 1 & \sin(\pi/3) & \sin(2\pi/3) & 2+\frac{\sqrt{3}}{4} \end{array}\right] \end{array} \Rightarrow \begin{array}{l} \alpha_0 = 2 \\ \alpha_1 = 0 \\ \alpha_2 = \frac{1}{2} \end{array}$$

b) Our interpolating polynomial is

$$p(x) = \alpha_1 + \alpha_2 \sin(x) + \alpha_3 \sin(2x) = 2 + \frac{1}{2}\sin(2x)$$

We can rewrite this as

$$p(x) = 2 + \frac{1}{2}\sin(2x) = 2 + \frac{1}{2}\, 2\sin(x)\cos(x) = 2 + \sin(x)\cos(x) = f(x)$$